

development is one area where it really helps not to have an additional cost for hardware or software maintenance. Even the other options TutorVista considered were open source—Python's Django, Ruby or Java.

TutorVista had to develop a white-board as there was no alternative. Ramasubramani reports, "We use tools that are highly specific to our area of work—and back then, there were no tools that we could have just purchased and used. We had to develop the white-board ourselves. When you build a tool from scratch, you can customise it exactly to your needs. Another option is to pick an already available tool, but it might not serve your purpose completely. In such circumstances, our team does a gap analysis, and we either change our scope and requirements so that they are met by the tool, or we change the tool itself to close the gap in features, or even request the developer community to build on it. At the end of the day, the business requirements should be met. It does not matter if you are using open source or proprietary tools, so long as you ensure your business need is satisfied. With proprietary solutions, it would not be as customisable, or we would have to pay a large sum of money to the implementation team. They would build it for you, and in all likelihood, they will hold the copyright. Total cost of ownership is not spoken about a lot, and is underestimated. When you're not paying maintenance fees year after year, and when you get what you want, it effectively translates to zero cost."

Apart from total ownership, the fact that there are tools already available in the open source domain, keeps TutorVista on the open source track. Ramasubramani feels this is one of the main reasons for the increase in the use of open source at an enterprise level. He illustrates his point, "If an educational company is setting up a learning environment system, it can buy proprietary tools like Blackboard, or it can use Moodle, which is open source. Moodle does not have all the features that Blackboard has, but open source developers are constantly adding more features to it. So overall, OSS is a win-win situation for all." Similarly, in the case of TutorVista, Jasper reports—an open source reporting engine—is being tested for reporting. "Rather than building reporting entirely from scratch, we are trying to see if we can use Jasper reports to help us create reports easily, without making the effort to build every report to the last level of details. We have a team that is analysing how we can make the best use of these reports in our day-to-day operations," Ramasubramani reveals.

He also feels that the organisation has saved a significant amount on costs by using open source software. Considering that the use of OSS means that there are no costs incurred for the software, for licences, maintenance fees, appropriate plug-ins to run

the platform, etc, the savings would add up to thousands of dollars.

The availability of tools and a sense of community are also some of the attractions of open source. Ramasubramani states, "We have not approached the community members for help yet, but we know that by going down this path, we can easily modify and reuse an open source tool with their help. That knowledge gives us the confidence that we do not have to build everything from scratch, but we can depend on the community for available open source tools and to develop or modify new ones."

The challenges down the road

Though it has been a smooth journey so far, Ramasubramani does admit that TutorVista still faces problems in sourcing personnel in certain key areas. "In terms of resources, we have a good team of people who know open source technologies. For instance, we always have had people who know how to set up a Linux server, handle Apache servers, and how to secure them. The only area where we have faced challenges is with regard to the MySQL database. MySQL is a great open source database in terms of what it can provide. The challenge, however, is in finding the right DBAs for it. If you consider databases like Oracle or SQL Server, they are supported by people who have been through well-established full-fledged training programmes. Since MySQL is more of a do-it-on-your-own mechanism, you will not find many DBAs lining up for certification in MySQL, unlike Oracle and MySQL."

Is open source for everyone?

Ramasubramani feels that it is not necessary that organisations will always save money by going the open source way. "It entirely depends upon the kind of business you are in. Start-ups would want to minimise costs in the right places. There are tools to do a cost-benefit analysis to check how you will benefit from a particular choice of software. When it comes to big enterprises—they are more monolithic, top-heavy, and not as agile as smaller companies. In such cases, proprietary tools and solutions may help. In cases where something goes wrong, you will have guaranteed support. Open source companies like Red Hat are also benefiting from this model. There are organisations that use Linux and sign up with Red Hat for support—they get the best of both worlds," Ramasubramani concludes. 

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